

Intro to Polyarthrititis and IMIDs

(immune mediated inflammatory diseases)

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LECOM

Objectives

- Present several common cases of “polyarthrititis”
- Discuss the general groups of polyarthrititis and their diagnoses:
 - Epidemiology
 - History and relevant ROS
 - Physical exam
 - Diagnostic tests
- Briefly discuss treatment options
- Basic immunology for clinicians

Immunology Basics



Case 1

- ID: 57 F with pain/stiffness in hands and fingers. She was diagnosed with “arthritis” 10 years prior to her first visit.
- PMH: htn, ‘dandruff’
- FH: brother with Crohn’s disease
- Meds: sulindac, norco

EXAM

- PE: swelling:
 - bilateral wrists
 - R thumb IP joint
 - Asymmetrical PIP's. No nodules
- Swollen right ankle and achilles tendon insertion.
- Fingernail changes



What tests do you want?

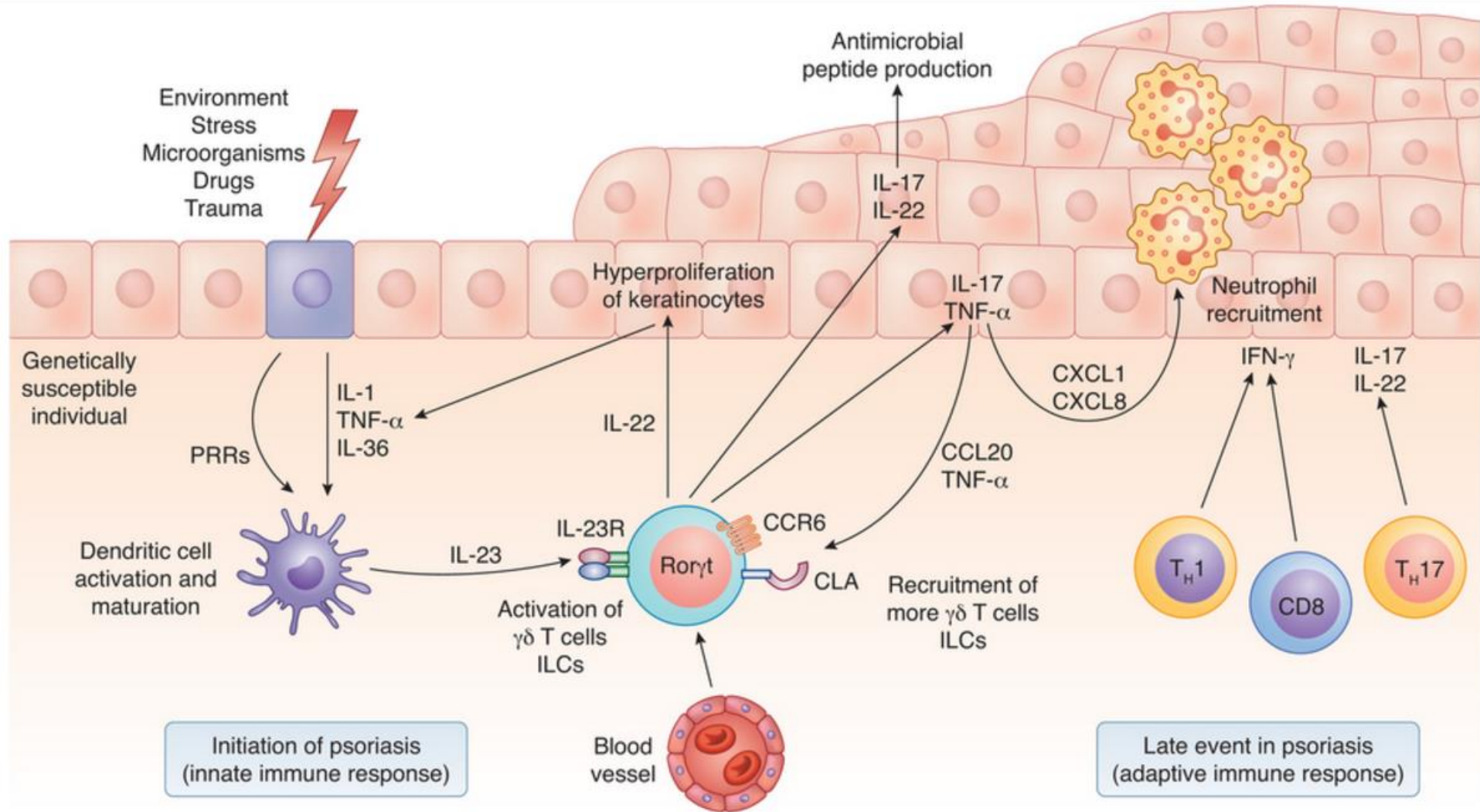
- A. Rheumatoid factor (RF)
- B. Anti-nuclear antibody (ANA)
- C. Anti-cyclic citrullinated peptide (CCP)
- D. Angiotensin converting enzyme (ACE)
- E. HLA-B27

Psoriatic Arthritis

- U.S. prevalence is 0.3 – 1% of population
- Arthritis occurs in ~20-35% of those with psoriasis
- M = F (RA is F>M)
- Onset: 30-50 yrs
- Unknown etiology of polygenic origin
 - HLA-B27
 - trauma
 - FH of spondyloarthritides
 - Ankylosing spondylitis
 - IBD (crohn's, ulcerative colitis)
 - Reactive arthritis

Psoriatic arthritis/Psoriasis

- ~2-3% of population have psoriasis
- >70% develop rash prior to arthritis
- ~15% develop arthritis concomitantly
- Rare cases of PsA without rash (future?)
- Pearls:
 - “Ray phenomenon”
 - Psoriasis is a risk factor for gout



Inflammatory Enthesitis



- Periosteal new bone formation
- Subchondral bone inflammation and resorption

Enthesitis



McGonagle D. *Arthritis Rheum.* 1999;42:1060-1066.

PsA Domains



- Peripheral arthritis
- Dactylitis/Enthesitis
- Axial disease
- Skin domain
 - Nail disease
- Arthritis mutilans



Taylor, W. Classification criteria for psoriatic arthritis: development of new criteria from a large international study. *Arthritis Rheum.* 2006 Aug;54(8):2665-73.



Psoriatic Arthritis Treatment

- NSAIDs
- Steroids (topical, IA)
- DMARDs
 - Leflunomide
 - Sulfasalazine
 - Methotrexate
- Others:
 - Ustekinumab/Stelara (IL-12/23)
 - Apremilast/Otezla (PDE-4)
 - Secukinumab/Cosentyx (IL-17)
 - Ixekizumab/Taltz (IL-17)
 - Abatacept/Orencia (T cell costim)
 - Tofacitinib/Xeljanz (JAK)
 - Upadacitinib/Rinvoq (JAK)
 - Guselkumab/Tremfya (IL-23)
 - Risankizumab/Skyrizi (IL-23)
 - Bimekizumab/Bimzelx (IL17A,F)
- Tumor necrosis factor inhibitors
 - Etanercept/Enbrel
 - Infliximab/Remicade
 - Adalimumab/Humira
 - Certolizumab/Cimzia
 - Golimumab/Simponi
 - Biosimilars









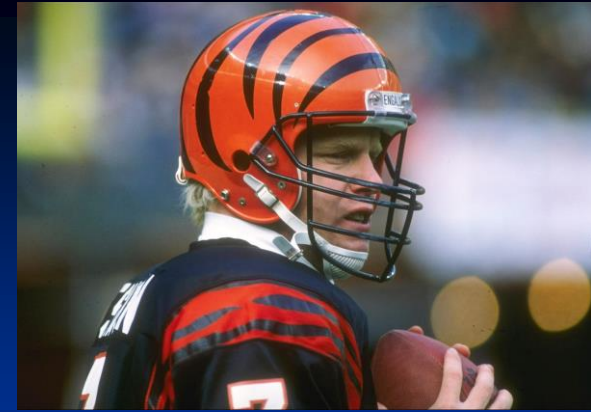
Questions about PsA?

■ Pearl....



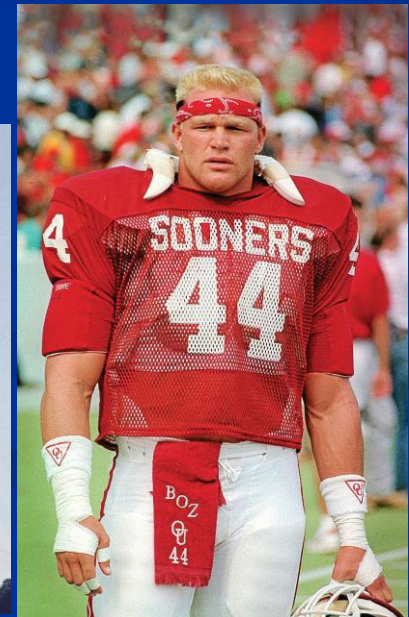
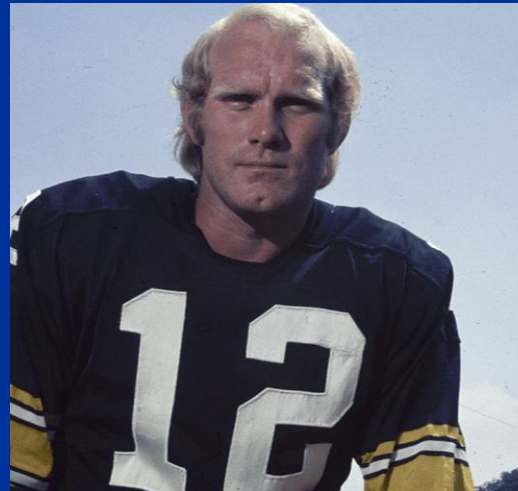


Case 2



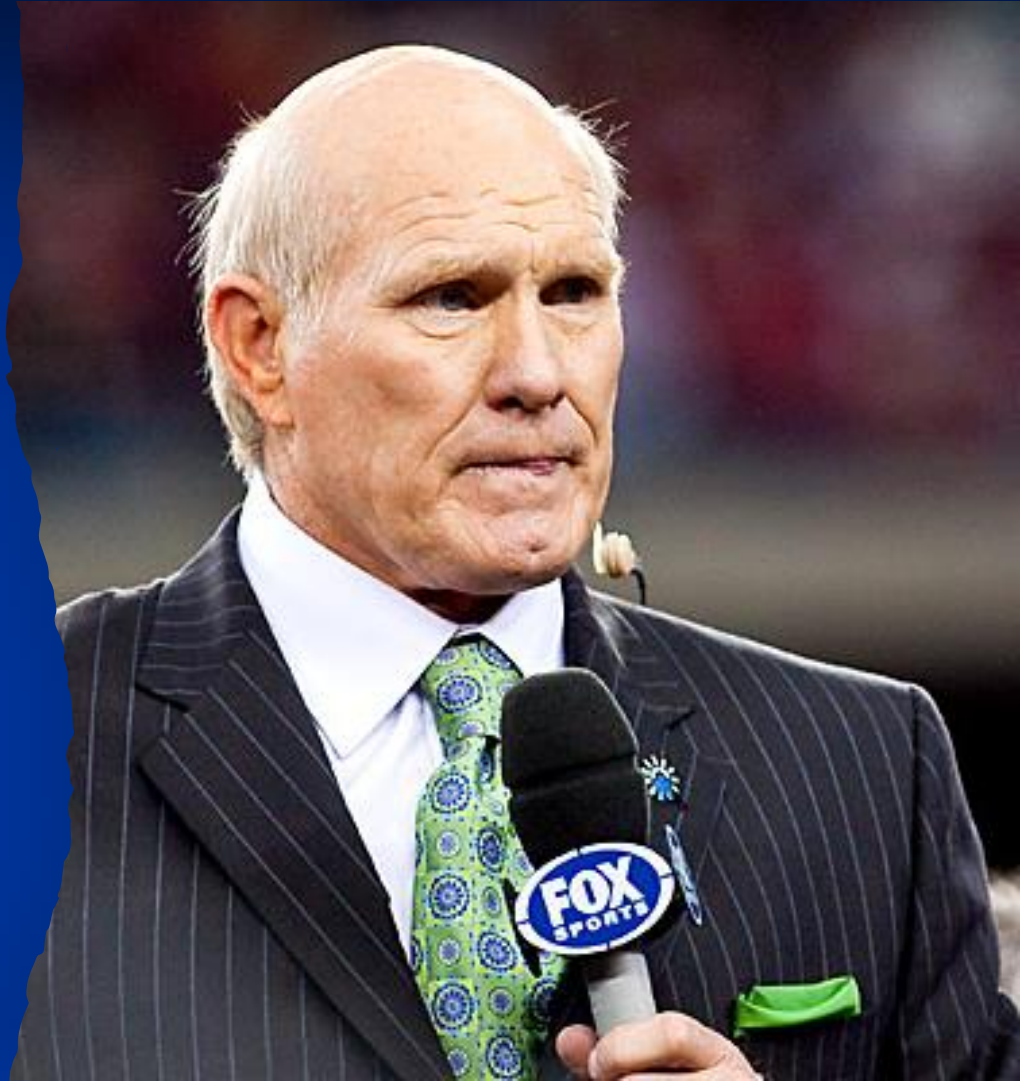
This former NFL star entered the broadcaster's booth soon after his playing days and remains an influential media voice but also serves as a patient and spokesman for the American College of Rheumatology after his inflammatory diagnosis:

- Boomer Esiason
- Terry Bradshaw ✓
- Cris Collinsworth
- Phil Simms
- Brian Bosworth



Diagnosis?

- ✓ ■ Rheumatoid arthritis
- Osteoarthritis
- Polymyalgia rheumatica
- Alopecia areata
- Acromegaly





Former Pittsburgh Steelers quarterback and rheumatoid arthritis patient Terry Bradshaw will serve as the ACR's official spokesperson for Rheumatic Disease Awareness Month in September.

<http://blog.arthritis.org/stories-of-yes/terry-bradshaw-rheumatoid-arthritis/>

<https://www.the-rheumatologist.org/article/hundreds-diseases-one-voice/>

Case 3

- ID: 56F referred: “suspected RA”; fibromyalgia, rash.
- PMH: OA, peripheral neuropathy
- Symptoms:
 - Myalgia
 - Joint pains/stiffness in MCP, DIP, PIP
 - no joint swelling
- SH:
 - Tobacco 1 ppd x 30 yrs
 - moderate etoh use
 - distant heroin use



What tests do you want?

■ Labs:

- ESR 6, CRP mildly elevated, RF 400
- Chemistry panel normal but
 - Albumin 3.0
 - AST/ALT: 90/100
- HIV(-)

■ Xrays:

- Feet and hands – mild osteoarthritis

■ Others:

- Anti-ccp negative, HCV positive
- + cryoglobulins

HCV

- 2.7-4.0 million infected in U.S.
- Rheumatic manifestations:
 - RF(+) high titre
 - Viral arthritis, myalgia
 - Mixed cryoglobulinemia
 - arthritis
 - palpable purpura
 - renal insufficiency
 - vasculitis

Mixed Cryoglobulinemia

■ CRYOGLOBULINS:

- Antibodies that precipitate in cool temperatures
 - In response to infection, malignancy, autoimmune disease

■ Clinically:

- anemia
- migratory arthritis
- ↓ GFR, hematuria, proteinuria
- rash
- hypocomplementemia

Other Viruses causing Arthritis

- Retroviruses: HIV, HTLV
- Chikungunya: East Africa, Seychelles, Mauritius
- Coxsackie A9, B2,3,4,6
- ParvoB19



“COVID TOES”



Treatment for Cryoglobulinemia

- Treatment for HCV
 - interferon/ribavirin
 - ledipasvir/sofosbuvir (Harvoni)
 - velpatasvir/sofosbuvir (Epclusa)
 - glecaprevir-pibrentasvir (Mavyret)
- Corticosteroids
- Plasmapheresis
- Cyclophosphamide
- Rituximab (anti-CD20 mAb)

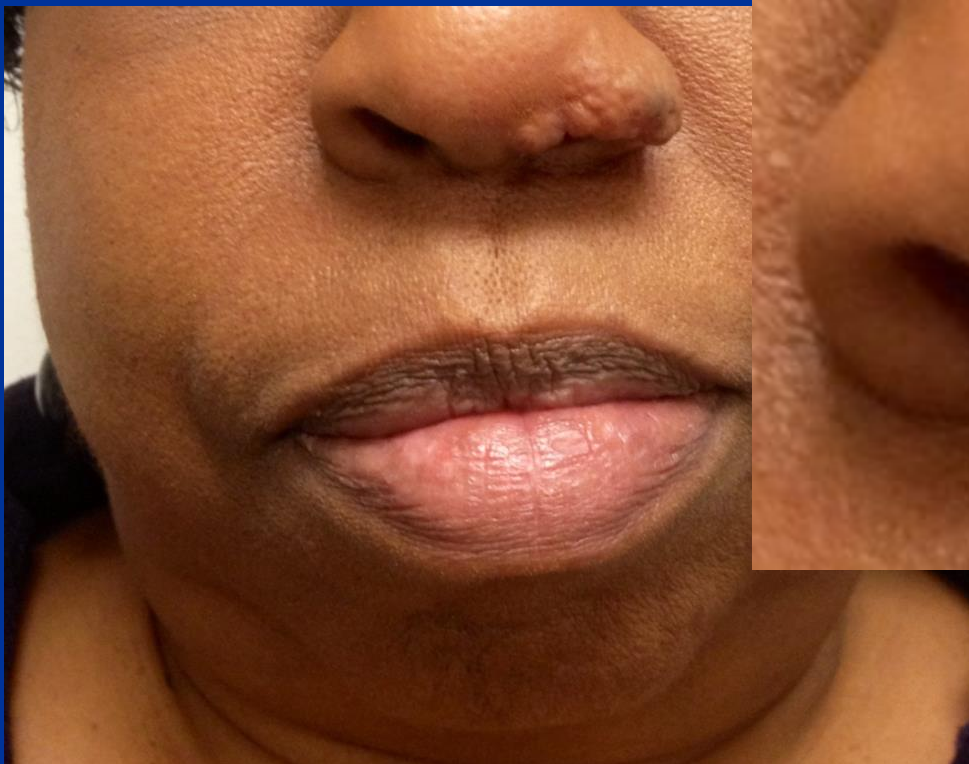
Questions on HCV, RF, cryo?

Case 4

- HPI: 48 yo AA female with 2 months of worsening swelling in bilateral wrists and scaly rash on nose. no recent travel
 - PMH: asthma
 - Meds: fluticasone, montelukast
 - SH: mod tob use x 20 yrs
 - FH: adopted

■ Physical exam:

- Mild bibasilar wheezes
- Mod wrist synovitis but no MCP swelling
- Skin:



Ex: Sep 12 2008

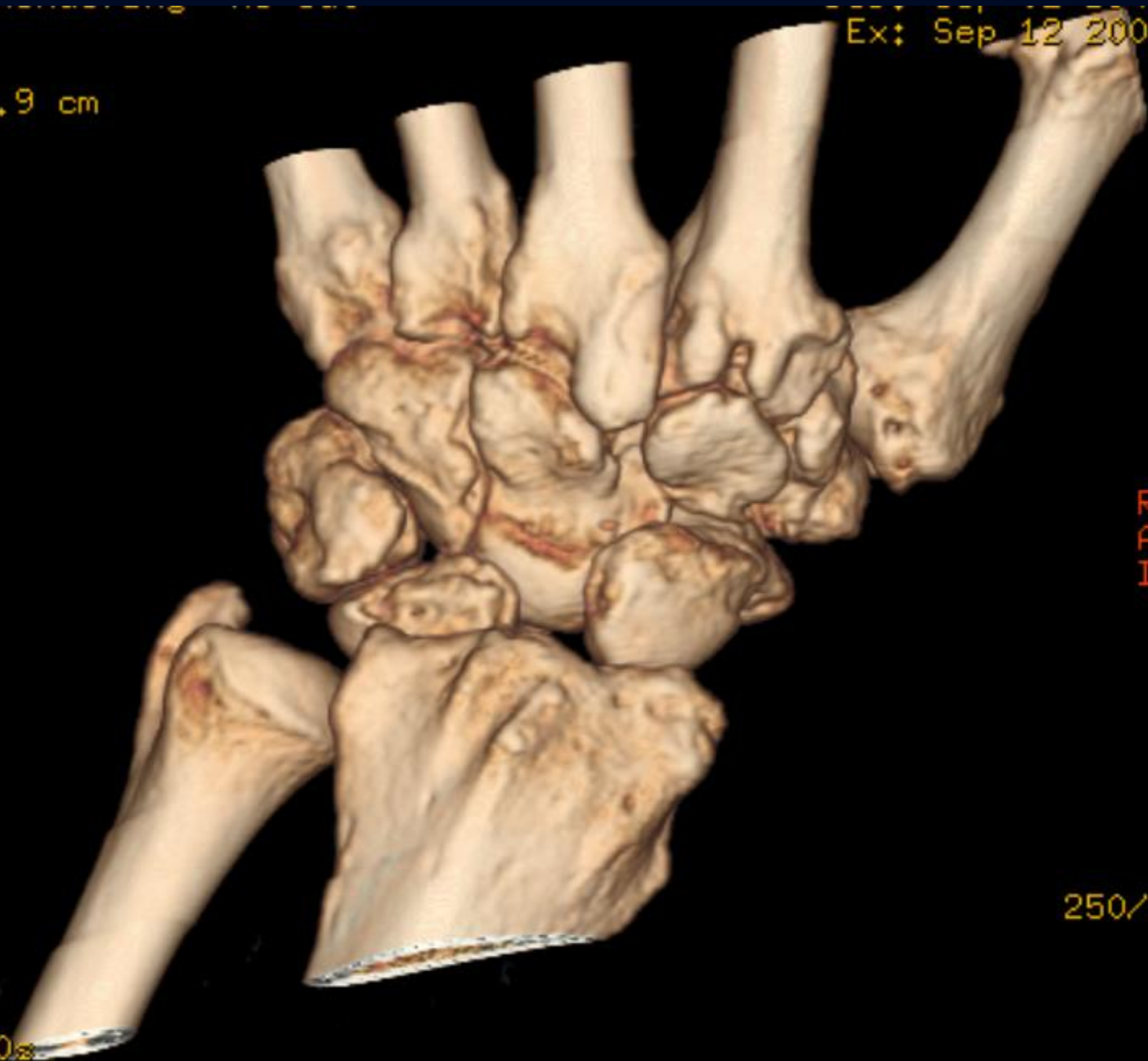
DFOV 12.9 cm
DETAIL

L
P
S

R
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No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/2



Ex: Sep 12 2008

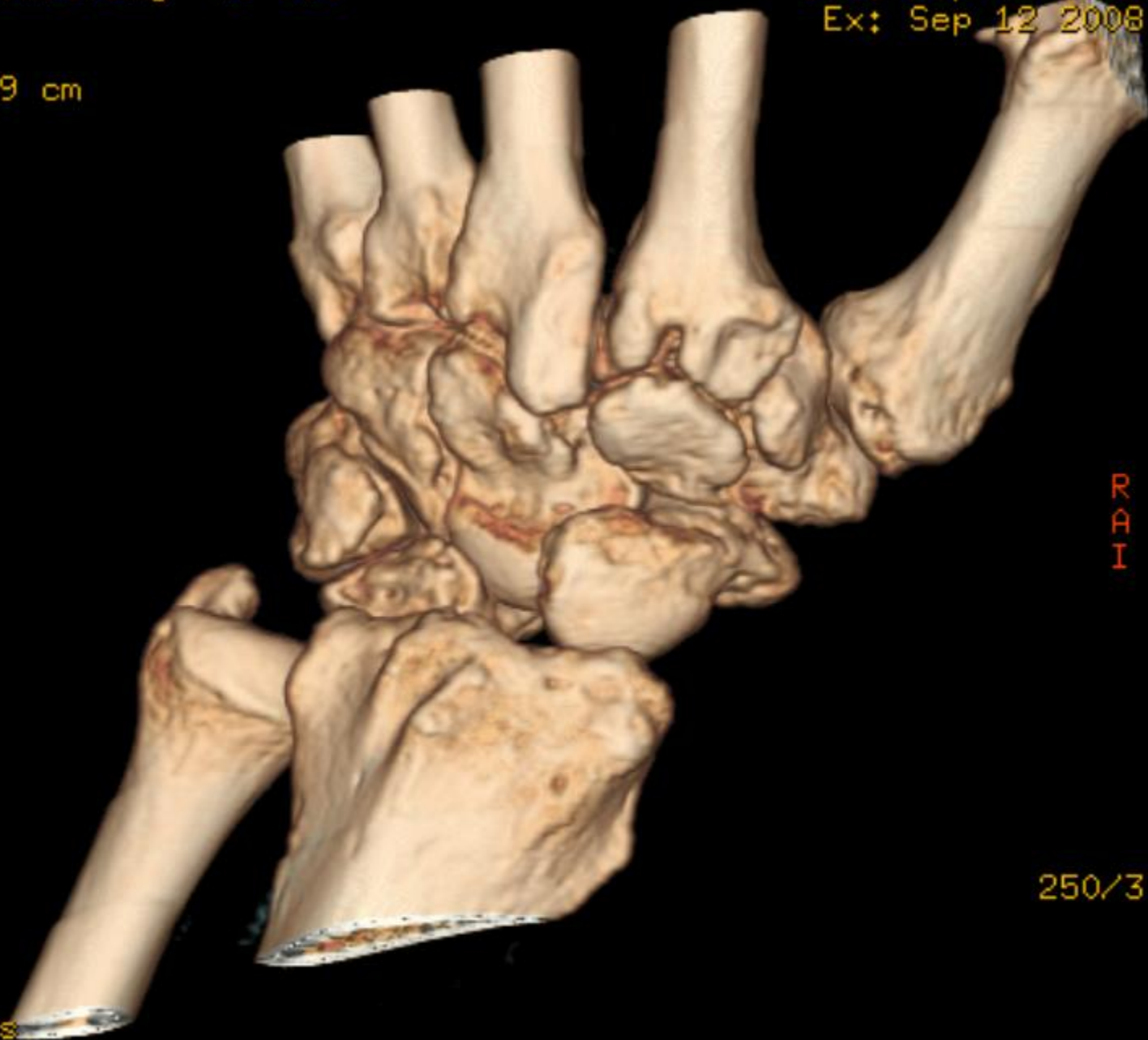
DFOV 12.9 cm
DETAIL

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No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/3



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

P
L
S

A
R
I

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/4



Ex: 5 Sep 12 2008

DFOV 12.9 cm
DETAIL

P
L
I

A
R
S

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/5



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

P
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R

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S
L

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/6

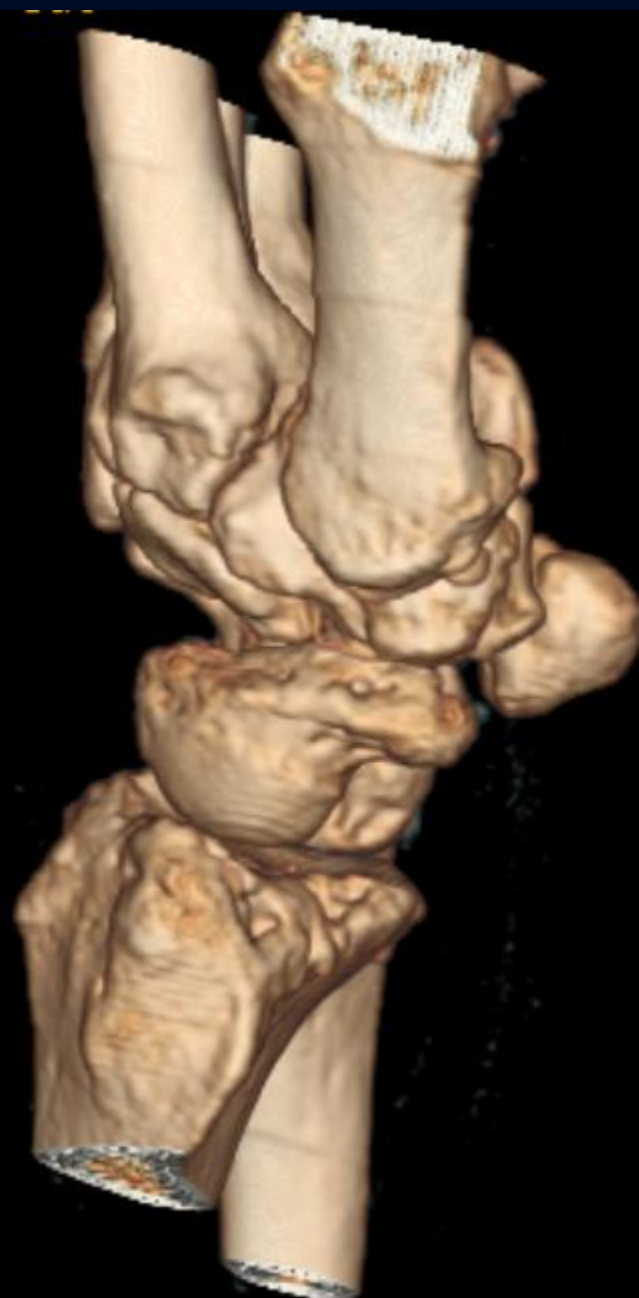


Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

P
R
I

A
L
S



No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/7

Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

P
R
I

A
L
S



No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/8

Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

R
P
I

L
A
S



No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/9

Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

R
I
P

L
S
A

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/10



Ex: Sep 12 2008

DFOW 12.9 cm
DETAIL

R
I
A

L
S
P

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/11



Ex: Sep 12 2008

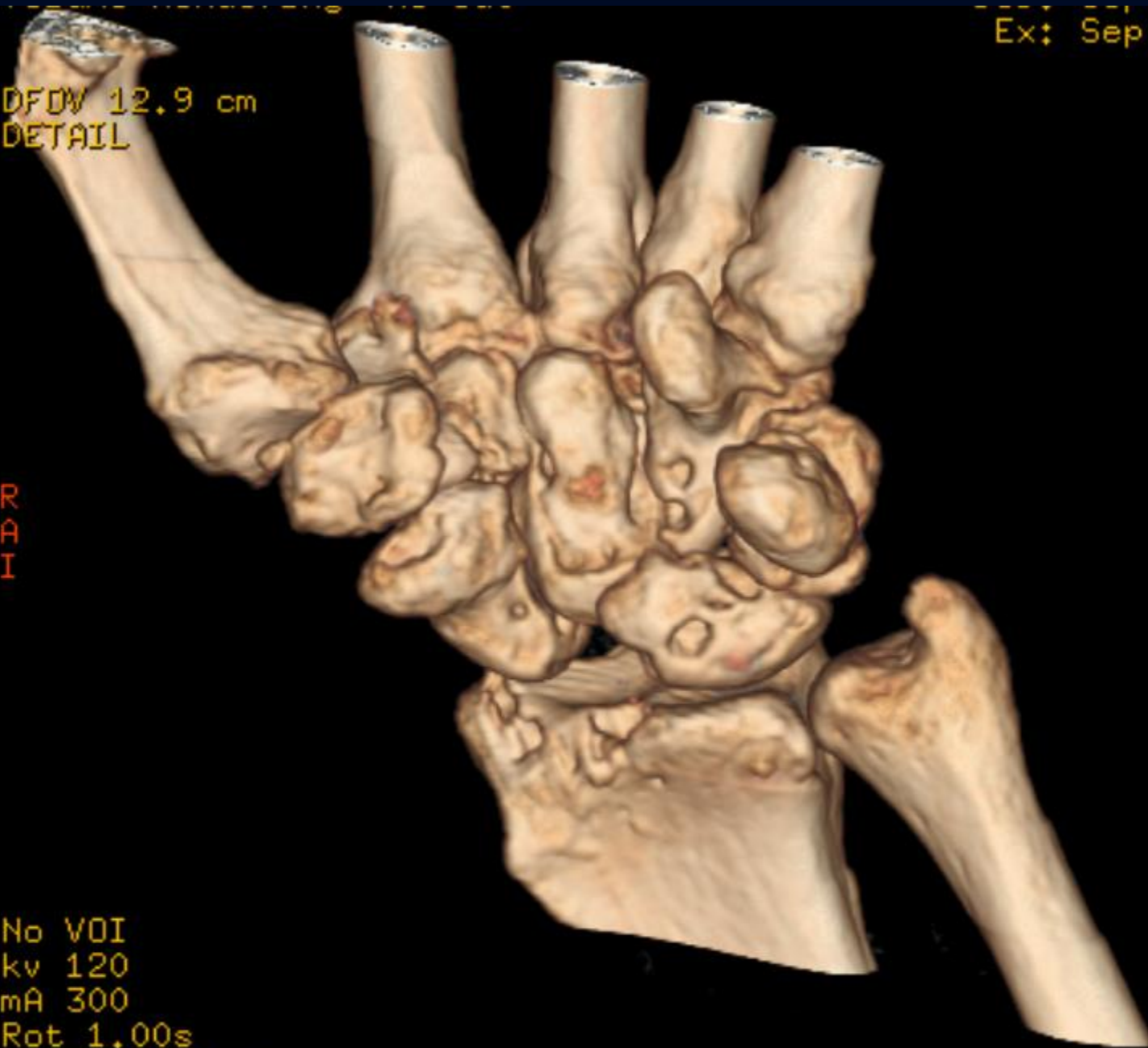
DFDW 12.9 cm
DETAIL

R
A
I

L
P
S

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/12



Ex: Sep 12 2008

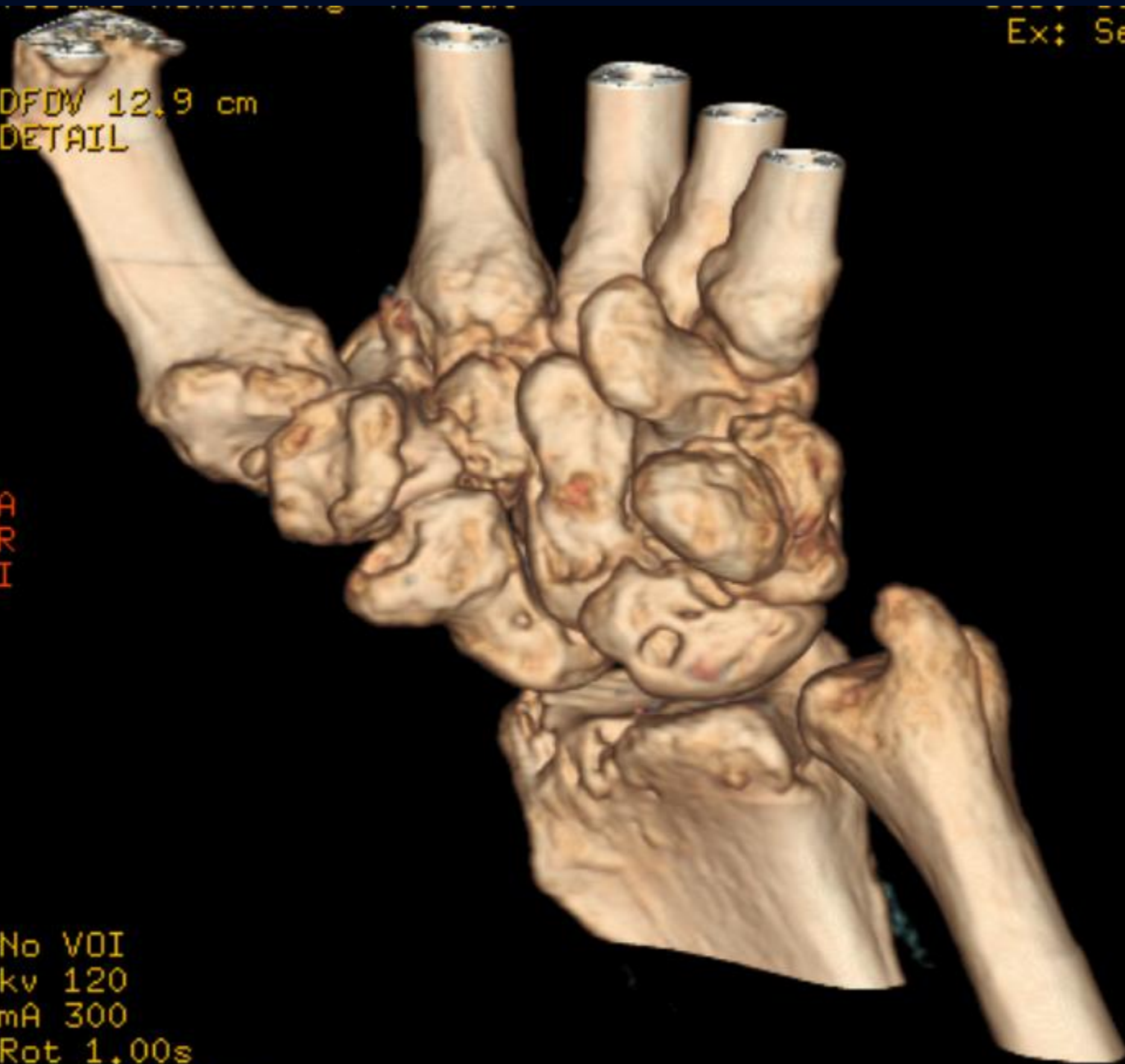
DFDW 12.9 cm
DETAIL

A
R
I

P
L
S

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/13



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

A
R
S

P
L
I

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/14



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

A
R
S

P
L
I

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/15



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

A
S
L

P
I
R

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/16



Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

A
L
S

P
R
I



No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/17

Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

L
A
S

R
P
I



No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/18

Ex: Sep 12 2008

DFOV 12.9 cm
DETAIL

L
A
S

R
P
I

No VOI
kv 120
mA 300
Rot 1.00s
0.6mm /0.6sp
Tilt: 0.0

250/19

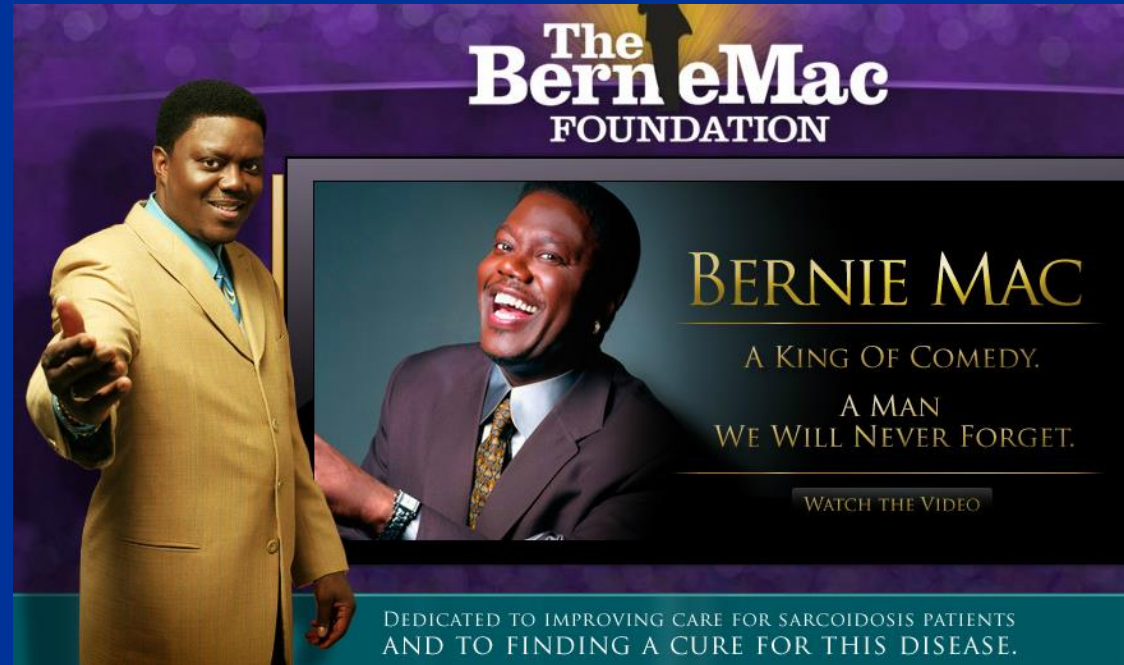


What is the best next test?

- A. ANA
- B. Serum amyloid
- C. ANCA
- D. Cocci titre
- E. ACE

Sarcoidosis

- General:
 - Heterogeneous, Idiopathic – genes? envt?
 - Non-caseating granulomas
- Epidemiology
 - Northern European, African
- Clinically
 - 90% with lung domain
 - Eyes, liver, skin, CNS
 - Ca, ACE nonspecific



Sarcoidosis Cases

■ LÖFGREN'S syndrome

- Hilar Lymphadenopathy
- Erythema Nodosum
- Arthritis



■ LUPUS PERNIO

- Misnomer

■ Sarcoid arthritis

- RA mimic
- Erosive granulomas

■ Treatments

- Corticosteroids
- Methotrexate
- TNF α inhibitors
 - infliximab

Questions about sarcoidosis?

Case 5

- ID: 69 F with aching hands and muscle aches in shoulders, arms. 10 min AM stiffness and intermittent swelling. Better in AM, worse by PM with use
- PMH: hypercholesterolemia, anemia, cataracts
- FH: daughter with OA age 50
- Meds: celebrex, tramadol

Case 5

- PE: Pt is 5'1", 165 pounds.
 - Bony nodular swelling at DIP and PIP
 - Valgus knees with mild effusions and crepitus.
 - Thin musculature in thighs
 - Tender in trapezius



© ACR

What is the most appropriate lab?

- A. Uric acid
- B. ESR
- C. CRP
- D. ANA
- E. No further lab testing

Osteoarthritis

- Most common joint condition
- Onset 40-50 years
- Joint space narrowing, osteophytosis
- Joints involved:
 - DIPs, PIPs, CMCs
 - Cervical, lumbar spine
 - 1st MTPs, knees, hips
 - **NOT shoulders, wrists or MCP's**

Fibromyalgia

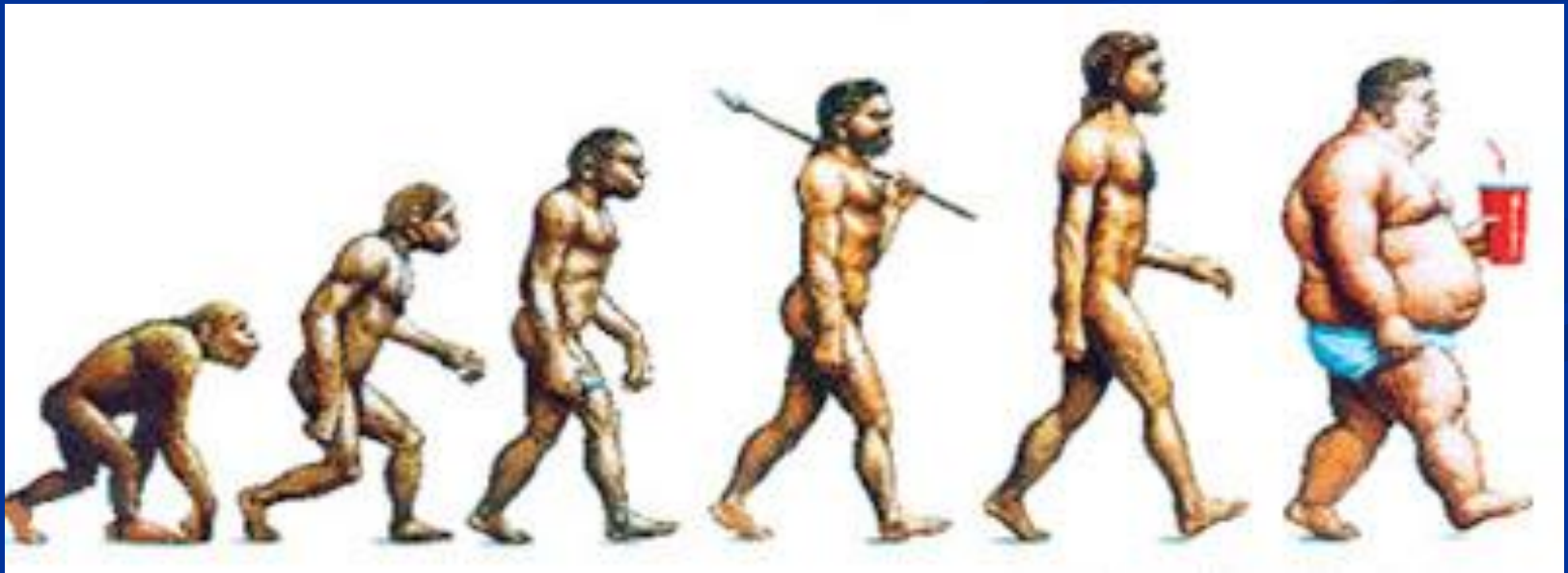
- Soft tissue myalgia
- Associated with other rheumatic conditions
- Sleep disturbances, depression, stressors, deconditioning
- Pathophysiology
 - Central sensitization
 - Allodynia
 - Nerve signals; ‘volume control’

OA contributing factors



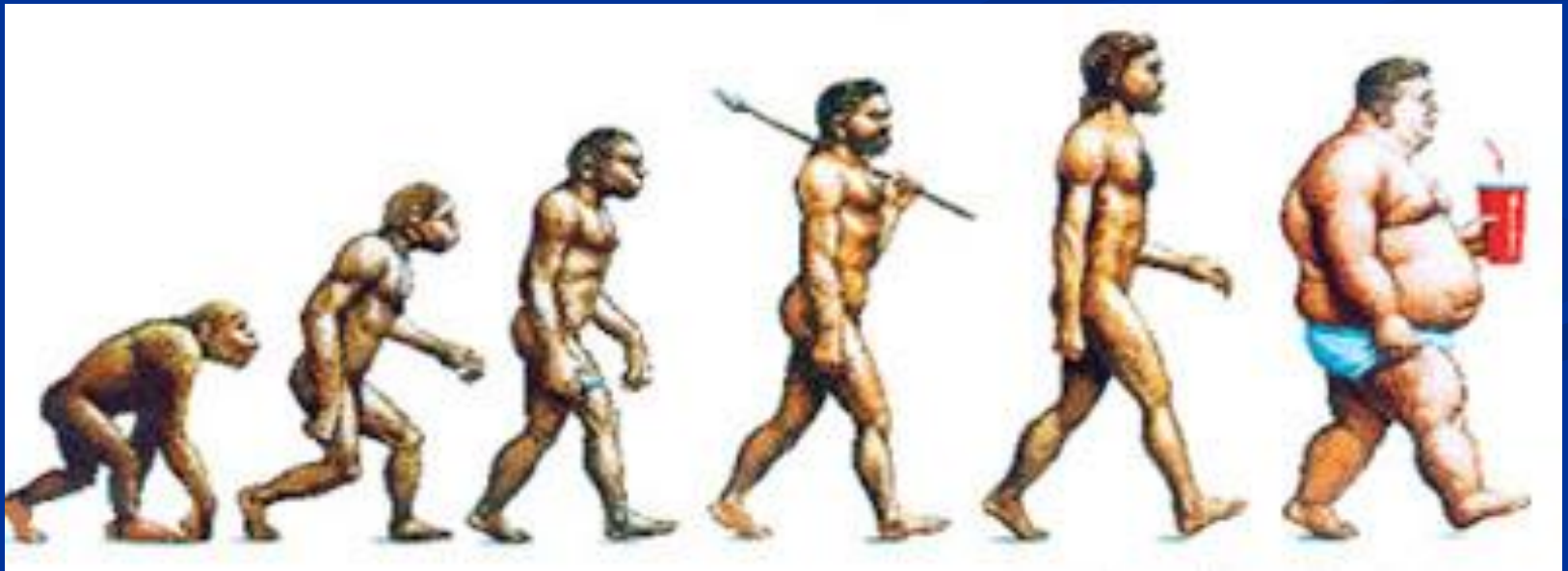
OA contributing factors

- Obesity



OA contributing factors

- Obesity
- Prior trauma and injury
- Menopause: estrogen?
- Family history



OA

■ Pathophysiology

- Periosteal bone remodeling (bone morphogenetic proteins) - subchondral microfractures
- Cartilage degradation (matrix metalloproteinases)



- Exam: bony enlargement, decreased ROM, peri-articular muscle weakness, crepitus

OA/fibromyalgia treatment

■ Non-pharmacologic:

- wt loss, PT, muscle strengthening, aerobic exercise
- Education; Integrative Medicine

■ Pharmacologic:

- NSAIDs
- Analgesia (tylenol, tramadol, opiates)
- Anti-depressants (nortriptyline, duloxetine)
- Anti-epileptics (fibromyalgia)

■ Injections of steroids (?) or hyaluronic acid (?)

■ Future?

- Anti-nerve growth factor
- Anti-MMP, anti-BMP



THE UNIVERSITY OF ARIZONA

Andrew Weil Center
for Integrative Medicine

Conclusions

- History and PE: recognition of msk syndromes:
 - duration of symptoms
 - pattern of joint involvement
 - family pedigree
 - comorbidities/ROS
 - associated rashes/systemic processes
- Laboratory findings and xrays

